

Ganesh Gupta

817-822-1131 | iamgs10rk@gmail.com | linkedin.com/in/ganeshhgupta | github.com/ganeshhgupta

EXPERIENCE

Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Built distributed search and data retrieval infrastructure using Elasticsearch with Knowledge Graph enrichment, engineering a pluggable pipeline executor routing nodes via typed REST APIs at sub-200ms p99 latency.
- Engineered a cloud-native executor routing pipeline nodes across Docker containers and AWS Lambda, running on both AWS and GCP (Cloud Run) with OAuth-based auth and RBAC for secure service access.
- Debugged and optimized backend services in Python and Java under production load, resolving latency regressions and data consistency issues across a distributed microservices fleet.
- Built observability for production inference infrastructure using Prometheus and Grafana, maintaining CI/CD pipelines with automated quality gates and rollback across Terraform-provisioned AWS/GCP infrastructure.

QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Cut p99 serving latency 420ms to 140ms across 50K+ daily requests by migrating synchronous services to async on GCP Cloud Run with Kubernetes-managed autoscaling.
- Operated and tuned a production vector/search retrieval stack (Pinecone, hybrid dense-sparse retrieval, re-ranking), improving recall 61% to 89% on a 2K-query eval set.

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Architected and operated 25+ production Java microservices on AWS (EC2, RDS, SQS) for 1M+ users at 99.9% uptime over 4 years, debugging distributed data consistency and performance issues in a regulated environment.
- Standardized REST API failure-handling contracts across 25+ Java microservices, reducing production escalations 60% via consistent retry, timeout, and circuit-breaker patterns.

PROJECTS

LangFetch

tinyurl.com/LangFetch

- Agent decomposing natural-language queries into dependency-ordered SQL subqueries with Elasticsearch/Pinecone-backed retrieval and hybrid re-ranking; deployed as an API server for enterprise data queries.

Distributed KV Store / VPC Control Plane Simulator

github.com/ganeshhgupta/distributed-kv-store

- Raft-consensus distributed storage engine handling concurrent writes with linearizable reads and log compaction; companion VPC control plane simulator modeling subnet routing across nodes.

Watch-your-LLM

github.com/ganeshhgupta/watch-your-llm

- Production observability platform tracking drift, latency, and regression alerting across deployments; FastAPI backend, PostgreSQL, Docker-based deployment.

EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Jul 2016 – Jul 2020

SKILLS

Languages: Java, Python, SQL, TypeScript, C++

Search & Distributed Systems: Elasticsearch, Vector Databases (Pinecone), Distributed Storage, Raft Consensus, Hybrid Retrieval & Re-Ranking

Cloud & Infra: AWS (EC2, Lambda, RDS, SQS, SageMaker), GCP (Cloud Run, Vertex AI), Terraform, Docker, Kubernetes, CI/CD (GitHub Actions)

Observability: Prometheus, Grafana, Production Monitoring, Drift/Regression Detection, Incident Response

CERTIFICATIONS

AWS Certified Developer Associate | Google Cloud Professional Machine Learning Engineer